

Prem Babu Kanaparthi

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SUMMARY

Inference Engineer specializing in low-latency LLM serving, multi-model routing, and cost-efficient production systems. Experienced in deploying scalable AI infrastructure on GCP/AWS with strong focus on latency, reliability, performance.

EDUCATION

Rochester Institute of Technology

Master of Science in Artificial Intelligence

August 2024 – May 2026

Rochester, NY

National Institute of Technology, Silchar (NIT)

Bachelor of Technology in Computer Science and Engineering

August 2020 – May 2024

India

TECHNICAL SKILLS

Backend & Programming: Python, SQL, FastAPI, REST APIs, PostgreSQL, MongoDB, PySpark

Distributed Systems & Data: Kafka, Streaming Pipelines, Feature Stores, Data Pipelines, Vector Databases, System Design

Cloud & Infrastructure: GCP, AWS (EC2, S3, Bedrock, SageMaker), Docker, Kubernetes, Terraform, CI/CD Pipelines, MLflow

Machine Learning: PyTorch, TensorFlow, Scikit-learn, LLMs, RAG, NLP, Computer Vision, Model Context Protocol

EXPERIENCE

Concentrix + Webhelp

February 2024 – July 2024

Generative AI Engineer

Newark, CA

- Designed and implemented a production multi-model LLM inference and routing system supporting 3+ foundation models, achieving 1.5s end-to-end latency, 18% reduction in inference cost, and stable performance across multiple enterprise applications.
- Built LiteLLM-based routing integrated with AWS Bedrock and SageMaker, handling thousands of daily inference requests while maintaining 95%+ response accuracy under varying latency and cost constraints
- Engineered end-to-end observability and evaluation pipelines using CloudWatch, introducing automated logging, safety checks, and drift monitoring, which reduced production incidents by 42% and improved mean time to detection (MTTD) by 35%.

AlphaBits Technologies

August 2023 – January 2024

Data Science Intern

India

- Redesigned ML experimentation, preprocessing, and evaluation pipelines across 5+ model variants, reducing model iteration time by 90% and improving ranking relevance by 10% on internal validation datasets.
- Implemented a centralized feature store and reusable training pipelines, enabling 100% reproducible experiments and supporting dozens of repeatable training and evaluation runs across ranking models.

iNeuron.ai

June 2023 – August 2023

ML Engineer Intern

India

- Developed supervised ML classification models for phishing domain detection using 20+ engineered URL and domain features, achieving 92% classification accuracy on held-out validation data.
- Integrated trained models into production-style services, adding logging, monitoring, and validation checks, ensuring consistent inference performance and enabling post-deployment performance tracking.

PUBLICATIONS

Lightweight Channel Attention for Efficient CNNs [paper] – Designed and evaluated a lightweight channel attention module (LCA) achieving competitive accuracy with negligible parameter and latency overhead on ResNet-18 and MobileNetV2.

PROJECTS

Emotion Engine – Emergent Appraisal in Generative Agents [Live Demo]

Feb 2026 – Apr 2026

- Built Emotion Engine extending Stanford Generative Agents with 72-feature LSTM that independently rediscovered OCC emotions (fear, grief, suspicion) from zero hardcoded rules ($p=3.3e-113$, $N=205K$ simulations), validated against Gallup 2024 baseline

Sushi – Serve Your Raw Data Perfectly [Website]

Feb 2025 – Apr 2025

- Built Sushi to automate repetitive data science workflows, enabling instant statistical analysis, hypothesis testing, regression, and outlier detection across 6+ dataset formats. Engineered FastAPI backend with intelligent sampling and caching (95% faster analysis), replacing manual multi-hour EDA with scalable real-time insights.

PackAI – Multi-Agent AI Coding Orchestration Extension [Extension Link]

Apr 2025 – Jun 2025

- Built PackAI VS Code extension using NLP intent analysis and capability-based scoring to orchestrate Claude, Copilot, and Codex in parallel, solving rate-limit bottlenecks. Implemented DAG-based task planning and conflict resolution, improving code quality and reducing hallucinations through coordinated multi-agent execution.

CERTIFICATIONS

Machine Learning Specialization (Stanford) [Link]